Durgesh Yadav

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M.Sc. Economics student at IIT Kanpur with a strong foundation in quantitative methods, data-driven market analysis, and team leadership. Demonstrated ability to translate analytical insights into measurable outcomes, complemented by hands-on entrepreneurial experience in scaling operations, improving efficiency, and strategically diversifying a growing venture.

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 – Present	M.Sc. Economics	Indian Institute of Technology Kanpur	6.57
2020 – 2023	Bachelor of Arts	Kanpur University	56.2%
2018 – 2021	Diploma in Mechanical Engg.	Lucknow Polytechnic, Lucknow	76.65%
2018	Class XII (UP Board)	Adarsh Public School	80.6%
2016	Class X (UP Board)	Adarsh Public School	84.5%

Note: BA (2020–23) and Diploma (2018–21) were pursued concurrently.

Scholastic Achievements

- Secured **All India Rank (AIR) 270** in the Indian Air Force Entrance Examination. *Jan '21*
- Achieved **99+ percentile in Physics** in the (JEE) Mains Engineering Entrance Examination. *April '2023*
- Cleared **Bhabha Atomic Research Centre (BARC) Stage-1** Examination in Mechanical Engineering. *October '24*
- Secured **All India Rank (AIR) 563** in the GATE (Economics) Examination. *April '25*

Work Experience

Team Leader | SPZ Solutions *Aug 2023 – Jul 2025*

- Led a team of 8 delivering academic content solutions, consistently achieving **2,500+ problems/month** at **95%+ accuracy** – exceeding company benchmarks every quarter.
- Introduced a structured *'Problems Faced'* reporting system to capture recurring errors, enabling data-driven process improvements and reducing repeat mistakes by an estimated 25%.
- Maintained team motivation through performance-linked incentives and continuous quality monitoring, sustaining high output during peak demand periods.

Entrepreneurial Experience

Co-Founder | Organic Poultry Farming Venture *Feb 2023 – Present*

- Co-founded an organic poultry farming business and scaled the venture substantially through operational efficiency and strategic business expansion.
- Achieved **32% Return on Capital Employed (ROCE)** and **40% CAGR** in revenue, driven by operational efficiency and integrated production.
- Diversified operations by launching an **egg production facility** and a **poultry health centre** to hedge against the risk of seasonal disease and reduce third-party dependency.

Key Projects

Financial Stress Index for India | *Empirical Research Project* – Under Prof. Sanjiv Kumar *Aug 2026*

- Constructed a composite **India Financial Stress Index (FSI)** using **286 monthly observations (2001–2025)** across banking-sector beta, equity returns and volatility, debt-market spreads, and an Exchange Market Pressure Index.
- Standardized and aggregated financial-market indicators to quantify **time-varying financial stress**, identifying pronounced fluctuations and periods of heightened financial-market instability.
- Aggregated the FSI to a **quarterly frequency** and examined its relationship with real economic activity, obtaining a **0.142 correlation with GDP growth**.

Tools: Python, Pandas, NumPy, Matplotlib, Financial Econometrics, Time-Series Analysis, Index Construction

Dynamic Portfolio Optimization under Stress | *Independent Quantitative Finance Project* *Sept 2025*

- Developed a **regime-switching portfolio model** by classifying NIFTY 50 market conditions into low- and high-volatility states and estimating state-specific return distributions and transition probabilities.
- Formulated a **CRRA utility-based dynamic asset allocation problem** with risky equity and a risk-free asset, applying **Value Function Iteration (VFI)** to obtain wealth- and regime-dependent portfolio policies.
- Identified substantially higher return dispersion in high-volatility regimes (**1.45% vs. 0.60% daily standard deviation**) and persistent regime dynamics, with a **52.6% probability of remaining in the high-volatility state**.
- **Key Results:** $\sigma_L = 0.60\%$; $\sigma_H = 1.45\%$; $P(H \to H) = 52.6\%$

Tools: Python, NumPy, Pandas, Matplotlib, yFinance, Dynamic Programming, VFI, CRRA Utility

GARCH-Based VaR & Tail Risk Analysis | *Independent Quantitative Finance Project* *May 2026*

- Developed an **empirical market-risk framework** using **2,711 daily NIFTY 50 observations (2015–2026)**, modeling volatility clustering and time-varying risk through **GARCH(1,1)**.
- Estimated highly persistent volatility dynamics with $\alpha + \beta = 0.982$ and an approximate **37-trading-day volatility half-life**; implemented 95% and 99% GARCH-based **VaR and Expected Shortfall (ES)** with out-of-sample violation rates of **5.52% and 1.29%**.
- Compared Gaussian and **Student-t GARCH** specifications, reducing **AIC from 5678.90 to 5579.64** and estimating fat-tail parameter $\nu = 6.95$; quantified extreme downside risk with **95% ES** $= -1.69\%$ and **99% ES** $= -2.83\%$.

Tools: Python, Pandas, NumPy, Matplotlib, arch, GARCH, VaR, Expected Shortfall, Statistical Modeling

Credit Risk & Probability of Default Modeling | *Independent Quantitative Finance Project* *Apr 2026*

- Analyzed **255,347 loan records** across borrower demographics, income, loan characteristics, credit scores, employment history, and debt exposure to identify key drivers of loan default.
- Developed a **Probability of Default (PD)** framework using Logistic Regression and tree-based classification models, incorporating feature engineering and class-imbalance treatment.
- Achieved **75.4% ROC-AUC** and **38.0% KS statistic**, with the Logistic model identifying **69.9% of default cases** through probability-based risk classification.

Tools: Python, Scikit-learn, Pandas, NumPy, Statistical Modeling

Option Pricing, Implied Volatility & Greeks | *Independent Quantitative Finance Project* *Mar 2026*

- Implemented the **Black–Scholes model** to price European call and put options and quantified sensitivity to underlying price, volatility, maturity, and interest rates through the **five major Greeks**.
- Calibrated **implied volatility from NIFTY option prices** and evaluated the model against observed market prices using pricing errors and out-of-sample validation.
- Analyzed the **volatility smile/skew and option-risk exposures** across moneyness and maturities, and constructed Greek-based **delta-hedging strategies** for portfolio risk management.

Tools: Python, NumPy, Pandas, SciPy, Black–Scholes, Options Greeks, Numerical Optimizations

Sales & Profitability Analytics Dashboard | *Independent Data Analytics Project* *Nov 2025*

- Analyzed **10,000+ retail transactions** across products, customers, regions, and categories to identify key drivers of sales and profitability.
- Built a **Power BI star-schema model** with DAX measures for revenue, profit, margin, AOV, YoY growth, and product-level performance; developed interactive executive and profitability dashboards.
- Evaluated the **sales–profitability trade-off of discounting**, identifying high-revenue/low-margin products, regional profitability gaps, and opportunities for targeted pricing and promotion.

Tools: Power BI, Power Query, DAX, Data Modeling, Excel, Business Analytics

Energy Uncertainty & ERPT to Food Inflation | *Coursework Research Project (ECO613M)* *Oct 2025*

- Replicated a nonlinear panel-threshold framework using **735 quarterly observations across 15 countries (2010Q1–2022Q2)** to examine how global energy-market uncertainty affects exchange-rate pass-through to food inflation.
- Estimated an energy-uncertainty threshold of **35.94%** using bootstrap threshold testing, separating the sample into low- and high-uncertainty regimes and estimating regime-specific exchange-rate pass-through.
- The exchange-rate pass-through was **found to be statistically insignificant in the low-uncertainty regime (0.1024)** but became **negative and statistically significant at approximately 0.29** under high uncertainty, indicating substantially stronger transmission of exchange-rate movements to food prices.

Tools: Python, Pandas, Statsmodels, Panel Econometrics, Threshold Regression, Bootstrap Inference, Time-Series Analysis

Carbon Sequestration & Economic Valuation | *Coursework Research Project (MBA667)* *Aug 2026*

- Conducted a **primary field survey of 228 trees across 19 species** at Hall-8, IIT Kanpur, using circumference, height, species, and wood-density data to estimate tree-level biomass and carbon stocks.
- Applied the **Chave et al. (2014) allometric model** and biomass-to-carbon conversion to estimate **2.17 tonnes of total biomass, 1.02 tonnes of carbon stock, and 3.73 tonnes CO_2e stored** across the surveyed trees.
- Fitted a **Chapman–Richards growth model** to project subsequent-year circumference and biomass, estimating an annual CO_2 sequestration potential of **1.32 tonnes CO_2/year** and identifying species-level contributions to the local carbon sink.

Tools: Python, Pandas, NumPy, Excel/Google Sheets, Allometric Modelling, Chapman–Richards Growth Model, Data Visualization

KPI-Based Performance Evaluation System | SPZ Solutions Pvt. Ltd *Nov 2023*

- Designed a **multi-dimensional KPI framework** to evaluate Academic SME performance across productivity, quality, accuracy, rework, turnaround time, on-time completion, and productive hours.
- Built an **Excel/Google Sheets-based analytics system** integrating daily activity logs, KPI normalization, weighted scoring, employee-level summaries, weekly performance analysis, and a **Manager's Index** for performance ranking.
- Developed a **management dashboard** with performance bands and automated tracking, enabling data-driven identification of high performers, performance gaps, and operational bottlenecks.

Tools: Microsoft Excel, Google Sheets, KPI Analytics, Dashboarding, Data Validation, Conditional Formatting

Technical Skills

- **Programming:** Python (Pandas, NumPy, Matplotlib), C++, LaTeX
- **Tools & Software:** Microsoft office, Microsoft Excel (Advanced), Power BI, SQL, Google Sheets, Git, MATLAB
- **Analytical Skills:** Machine Learning (Regression, KNN, PCA, Neural Networks/LSTM), Time Series Analysis, Econometric Modeling, Data Visualization, Financial Analysis

Positions of Responsibility

- **Student Guide** – IIT Kanpur
  – Guided students on academic, administrative, and campus-related matters, facilitating their transition and access to university resources.
- **Volunteer Teacher** – Prayas, IIT Kanpur
  – Taught and mentored school students through **Prayas**, supporting their academic development through regular instruction and guidance.

Relevant Courses

Applied probability and statistics — Macroeconomics — Mathematical Methods for Economics — Quantitative Methods in Economics — International Economics — Issues of Economic Growth and Development — Game Theory Applications — Econometrics — Computational Methods in Economics — Economic Data Analysis: Methods and Computational Aspects — Advanced Topics in Micro Economics — Foundations of Financial risks — Sustainable Development For Business